BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 248 · Autumn mock feedback and intervention

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why must binary search use sorted input?

Q2. What is a base case?

Q3. Why do exam questions award method marks?

Q4. A rising-edge D flip-flop samples D=1 at the first clock edge. D then changes to 0 between edges, and another rising edge occurs. State Q immediately after the first edge, between edges, and after the second edge.

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I Do · Read new knowledge

Key vocabulary

diagnostic — Assessment used to locate a learning gap.

method mark — Credit for a correct intermediate step.

transfer — Applying a method to a changed problem.

correction — Justified replacement for incorrect reasoning.

New learning

Diagnose the weakness — Look for a pattern across the marked questions: omitted intermediate states, incorrect notation or missed explanations.

Rebuild the method — Model each calculation, comparison or state change instead of simply copying the final answer.

Test transfer — Give students a different array or bit pattern and check whether they can produce the same method independently.

Worked example

Correct an incomplete two's-complement answer for −9 in eight bits.

1. Predict first — Start with +9 in exactly eight bits and invert it.

2. Explain the answer — Add one to give 11110111, representing −9. This explicitly shows the stages.

We Do · Practise with guidance

A student writes only '6' for shortest distance A→C using A–B=2, B–C=4, A–C=9. Improve it.

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A→B→C costs 2+4=6.
Direct A→C costs 9.
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Predict first

Compare both candidate routes, recording their total costs.

Explain the answer

The two-edge route costs 6, which is smaller than the direct route cost 9; explain why.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Improve an answer to 'trace linear search for 7 in [3,5,7,9]' by showing comparisons and returned index.

Question 2

Correct the claim '−6 is 11111001 in eight-bit two's complement'. Provide the full method.

Question 3

A student traces binary search for 13 in [1,3,5,7,9,11,13] but does not change high or low after a comparison. Rewrite the trace with bounds and explain why it terminates.

Exit Ticket · Answer before checking

Copying only a mark-scheme answer proves you can repeat the method.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It discards half the candidates based on ordering.

Q2. A condition that stops recursive calls.

Q3. They credit correct intermediate reasoning and algorithm states.

Q4. Q=1 immediately after first rising edge; remains 1 between edges even though D changes; at next rising edge Q becomes 0.

You Do pages · Independent questions

Q1. [4] Compare 3 [1], 5 [1], 7 [1], then return index 2 (zero-based) [1].

Q2. [6] Positive 6=00000110 [2]; invert to 11111001 [2]; add one to 11111010 [2]. The claimed value is one less and represents −7.

Q3. [6] low0 high6 mid3→7 [1]; target larger so low4 [1]; low4 high6 mid5→11 [1]; increase low6 [1]; mid6→13 found [1]; bounds shrink on each unsuccessful step [1].

Exit Ticket · Model answer

False. Independent success with a different example is better evidence.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: